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RESEARCH ARTICLE

An Efficient Approach for the Detection and Prevention of Gray-Hole Attacks in VANETs

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ABSTRACT Vehicular Ad-Hoc Networks (VANETs) deliver a wide range of commercial as well as safety applications and further motivate the advancements of Internet of Vehicles (IoV), Intelligent Transportation Systems (ITS), and Vehicles to Everything (V2X) communication. Despite their potential benefits, VANETs are susceptible to a variety of security attacks due to their open, distributed, and dynamic nature, which includes intrinsic protocol design issues. One such an infamous security attack is the Gray-Hole Attack (GHA), typically has two variants: *Smart GHA* and *Sequence Number-based GHA*. In Smart GHA, the malicious node behaves normally during the route discovery process, while in Sequence Number-based GHA, the malicious node starts misbehaving during the route discovery process. In either case, once the route is successfully established, it starts dropping the packets. In this paper, a novel security approach called “Detection and Prevention of GHA” (DPGHA) is proposed to detect and prevent both variants of GHA in Ad Hoc On-Demand Distance Vector (AODV) based VANETs. The approach is based on the generation of dynamic threshold values of abnormal differences of received, forwarded, and generated control or data packets among nodes and their sequence numbers. The proposed DPGHA is implemented and tested in NS-2 and SUMO simulators and its various performances are compared with the most relevant benchmark approaches. The results showed that the proposed DPGHA performed better than the benchmark approaches in terms of reduced routing overhead by 10.85% and end-to-end delay by 3.85%, increased Packet Delivery Ratio (PDR) by 4.67% and throughput by 6.58%, and achieved a maximum detection rate of 2.3%.

INDEX TERMS AODV, GHA, OBU, RSU, VANET.

I. INTRODUCTION

Vehicular Ad Hoc Network (VANET) is a special type of ad hoc network in which a group of moving vehicles and fixed Road Side Units (RSUs) are linked together over a wireless medium to provide a safe and secure traffic environment [1], [2], [3]. Due to its utility, accessibility, and flexibility, new vehicles have been launched on the market with embedded onboard sensors that enable them to communicate with each other and exploit the benefits of VANET. Such systems are

best suited for a wide range of applications, including safety-related, traffic and fleet management, and entertainment [4].

From the architectural point of view, a VANET contains three types of components: OBUs (On-Board Units), RSUs, and Trusted Authority (TA). These components further yield three classes of communication modes: Vehicle-to-Vehicle (V2V), Vehicle-to-Infrastructure (V2I), and Infrastructure-to-Infrastructure (I2I) communications [5], [6]. VANET is the backbone of the ITS, which plays a crucial role in providing real-time and sensitive information to drivers, passengers, and traffic administrative authorities [7], [8].

Security Issues in VANET: Despite their outstanding applications, significance, and adaptability, VANETs possess

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several security issues and challenges due to their unique characteristics. For instance, the fast-moving, open access medium, highly dynamic network topology, frequent disconnection, large-scale networks, and protocol design issues have made the VANETs' communications more susceptible to numerous security attacks [2], [4], [9], [10]. In AODV-based VANET, two nodes can't share data without the help of intermediate nodes. Because the destination node is not often lying in the source node's transmission range. Thus, routing protocols are used to determine the route among nodes. Over time, several routing protocols have been developed. Out of these, AODV [11] is considered to be the most widely used routing protocol in VANETs [12], [13], [14].

The AODV routing protocol provides a faster, more dynamic network connection with minimum computations and storage requirements, making it an excellent alternative for the highly dynamic nature of VANET [14], [15]. However, there are several significant security challenges associated with the AODV routing protocol making VANET more prone to several attacks, including Black-Hole Attack (BHA), GHA, Wormhole Attack, etc. [4], [14], [16], [17], [18].

Motivation: Due to their unique properties and features, the applications and services of VANETs have constant security threats and attacks. One such challenging attack which compromises the security messages exchanged in VANETs is a GHA. It is a harmful attack in which the behavior of a malicious node is extremely unpredictable. GHA sometimes acts as an honest node, but afterward, it starts misbehaving by dropping the packets partially, selectively, or completely [19], [20]. Second, it sporadically drops the packets from a specific node while forwarding the packets to all other nodes [21], [22], [23]. These packets may carry critical safety-related messages and warning notifications. Furthermore, dropping these packets may lead to a decrease in the network's overall security and performance, resulting in traffic deaths, accidents, and congestion. Hence, the mitigation of GHA is a very crucial issue within the AODV-based VANET, because, compared to other attacks, the identification of GHA is extremely difficult, due to its hypocritical behavior. Although some attempts have been made in the relevant literature to address GHA. But, most of them have increased routing overhead and delay, while decreasing throughput and PDR.

Contributions: The work presented in this paper is an extension of our previous research study focused on the detection and prevention of black-hole attack [18]. Advancing it further, this work is contributing a new solution for the detection and prevention of gray-hole attack with the following notable contributions to the field.

- An efficient approach for the detection, as well as prevention of GHAs is proposed, to improve the security and overall performance of AODV-based VANETs.
- Unlike the existing approaches that were based on static thresholds, the proposed approach is based on the generation of dynamic thresholds of abnormal differences of

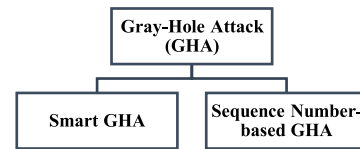


FIGURE 1. Types of GHA.

received, forwarded, and generated control/data packets and their sequence numbers.

- The proposed approach is dual-purpose (two-in-one) because it can detect and prevent both variants of GHA i.e., Smart GHA and Sequence Number-based GHA.

The rest of the paper is organized as follows: Section II briefly describes the background of GHA; Section III presents the related work; Section IV describes the proposed DPGHA; Section V explains the implementation and performance analysis; Finally, Section VI concludes the whole paper.

II. THE GRAY-HOLE ATTACK (GHA) IN VANETS

The behavior of nodes has a major impact on the overall security and performance of VANETs as compared to other wireless networks. Because it is very difficult for a node to determine whether the information it received came from a legitimate or malicious node. For example, in the AODV protocol, when a node wants to communicate with another node, a route discovery process is initiated by broadcasting the Route Request (RREQ) to all its neighboring nodes. Upon receiving the first Route Reply (RREP), a bidirectional path is established, and the source node starts transferring packets toward the destination. In AODV-based VANETs, the source node is always oblivious to the exact destination, making them more susceptible to many security attacks, including GHA.

GHA is a variation of BHA (in which a malicious node completely drops the packets) [18]. GHA sometimes behaves like a normal node while other times it acts maliciously by dropping the packets partially, selectively, or fully. Broadly, GHA can be classified into two types: Smart GHA and Sequence Number-based GHA [19], [20], as shown in Figure 1.

In the Smart GHA, the malicious node behaves normally during the route discovery process. However, once the route is successfully established and the data packets begin to transfer, it either starts dropping the packets partially, or fully, or forwards them honestly. On contrary, in Sequence Number-based GHA, the malicious node sends a fake RREP with a higher sequence number in response to the RREQ in order to attract the flow of packets towards itself. In such a case, once the route is established it completely drops the packets. The following is a list of properties that a gray-hole node (either a Smart or a Sequence Number-based GHA) adopts during communication with other nodes in the AODV-based VANET.

- The number of RREPs (in response to RREQs) generated by a gray-hole node is very high.
- The number of RREQs generated by a gray-hole node is very low and often equal to zero.
- The number of data packets received by a gray-hole node is very high.
- The number of data packets forwarded by a gray-hole node is lower than its received packets.
- The RREP of a gray-hole node (sequence number based) contains a larger sequence number.
- Due to existing of GHA in a VANET, the PDR of such a network is significantly decreased while the Packet Loss Ratio (PLR) is vividly increased.

Figure 2 (a-e) illustrates various scenarios of GHA. Here we assume that vehicle V_1 is the source node, V_5 is the destination node, V_2 is a gray-hole node, and all the rest are normal intermediate nodes. In Figure 2 (a), source vehicle V_1 initiates a routing discovery process by broadcasting an RREQ packet to all its neighboring nodes for establishing a route toward the destination vehicle V_5 . Here, vehicle V_2 acts as a Smart GHA that participates normally and exchanges control packets to its neighboring nodes. Next, in Figure 2 (b), vehicle V_2 acts as a Sequence Number-based GHA by sending a fake RREP first with a higher Destination Sequence Number (DSN) in response to the RREQ of vehicle V_1 .

However, as shown in Figure 2 (c), once the route is established and the data packets begin to transfer towards destination V_5 , vehicle V_2 (as a Smart GHA) drops the packets partially. Similarly, in Figure 2 (d), vehicle V_2 forwards all the received packets from source vehicle V_1 , showing a normal node behavior. Finally, in Figure 2 (e), vehicle V_2 (maybe a Smart or Sequence Number-based GHA) completely drops all the packets received from source vehicle V_1 .

Figure 3 illustrates a visual representation of the impact of GHA in a VANET. In this scenario, vehicle V_4 (gray-hole vehicle) receives an accident alert message from V_3 but it does not forward it to the approaching vehicles, i.e., V_5 and V_6 . Instead of forwarding this important message to other vehicles, V_4 intentionally drops it, resulting in additional causes of more accidents and traffic jams.

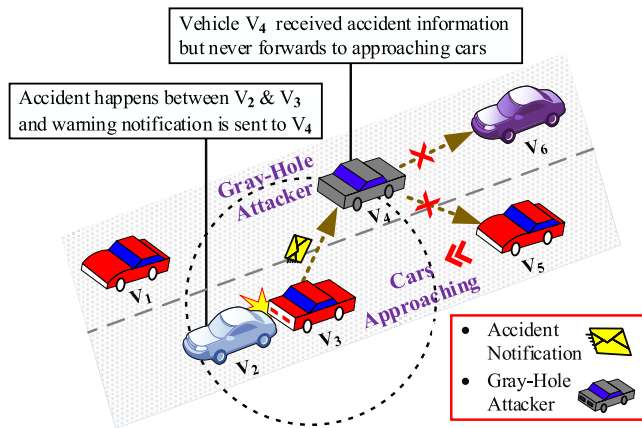
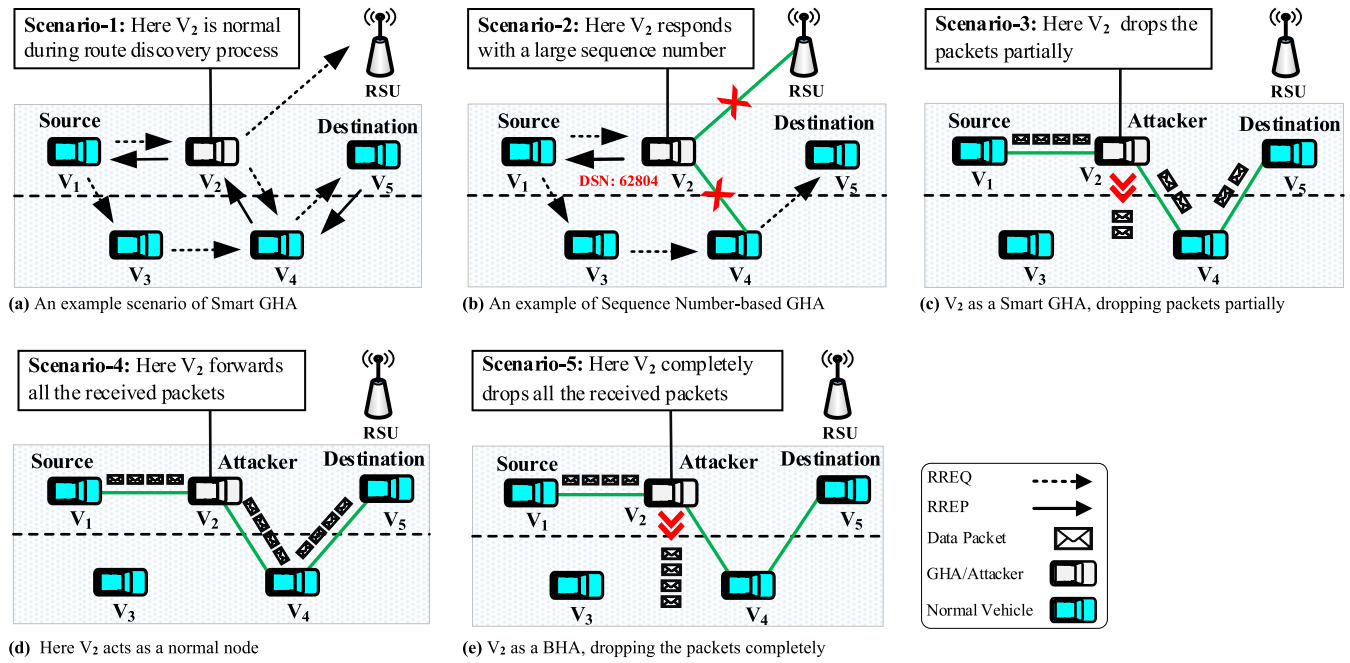
III. RELATED WORK

Over the time, many approaches have been proposed to ensure the security of VANET in general and to eradicate GHA in particular. One such notable approach is proposed by Su [15] an Anti-Blackhole Mechanism (ABM), based on the amount of anomalous discrepancy between RREQs and RREPs transmitted from a node. When the suspicious value of a node exceeds a certain level, the node sends a block message to inform all other nodes. However, this approach has several weaknesses, including latency in determining the malicious nodes, overhead in distributing revocation information across the network, and the generation of a large amount of Intrusion Detection System (IDS) traffic.

Khattak et al. [24] proposed a hybrid solution based on the assumption that a malicious node always sends the first RREP. The first RREP is ignored by the source node, and the second RREP is chosen for data transmission toward the destination. In this way, the likelihood of a malicious node on the second RREP's path is decreased, resulting in a safer route for transmission. However, this technique fails in the case when a malicious node is in the vicinity of the source node and the destination is far away. Likewise, an end-to-end delay will occur if the second path is selected, voiding the presence of any malicious node. Kadam et al. [25] proposed a DPMV (Detection and Prevention of Misbehaving Vehicles) approach based on caching mechanisms that improve the Detection of Malicious Vehicles (DMV) scheme presented in [26]. It first examines all the available routes for the presence of malicious nodes. If a route containing an attacker is discovered, the route is ignored, and a new route toward the destination is established. In comparison to DMV, this approach detects malicious nodes efficiently with high mobility. However, the approach requires more time for its processing, resulting in a high end-to-end delay [12].

Khan et al. [27] proposed a cluster-based scheme called DMN (Detection of Malicious Nodes) against packet-dropping attacks. In this scheme, when a fresh vehicle enters the proximity of a cluster, a verifier vehicle begins monitoring the behavior of the newly arrived vehicle. The verifier vehicle notifies the Cluster Head (CH) if it crosses the specified threshold level. After confirmation, the CH further reports it to the nearest TA where it is removed from the list of trusted nodes. The major drawback of this scheme is that it generates a huge routing overhead due continuous formation of clusters, maintenance, and exchange of information among nodes, thus reducing the overall efficiency of the VANET. Ali Alheeti et al. [28] proposed an anomaly-based IDS for GHA and rushing attacks. The IDS component is installed on each node which regularly monitors the behavior of other nodes. The IDS collects data from a trace file generated by a simulation environment to determine whether a vehicle's behavior is normal or malicious. However, monitoring every node through the IDS component leads to excessive processing overhead and end-to-end delay.

Wahab et al. [29] proposed the CEAP (Collection, Exchange, Analysis, and Propagation) model to detect packet-dropping attacks. To collect the data, the nodes are designated as watchdogs that continuously monitor the behavior of other nodes. Further, the Support Vector Machine (SVM) techniques are employed to analyze the collected evidence. Finally, the malicious nodes' information is shared among CHs through the propagation phase. However, the additional responsibility of IDS on the CH node has an impact on the performance of the CH as well as on the network. Tyagi et al. [22] proposed an ES-AODV (Enhanced Secure AODV) approach aimed to detect malicious nodes in VANETs. In this approach, an RSU serves an additional responsibility of the Certification Authority (CA). Vehicles are verified by an RSU before going to initiate any



communication. A node is detected as malicious when a node exceeds the threshold level of RREPs' sequence numbers and verifies the nonce value. However, the approach is based on cryptographic techniques, which offer protection against external security attacks. It may fail in a scenario where internal security attacks exist in a network.

Zardari et al. [30] proposed a DDBG (Dual-Attack Detection of Black-hole and Gray-hole Security Attacks) approach based on the generation of a status packet. In this model, the IDS node floods status packets into the network and then waits for their replies. Upon receiving a status packet, each legitimate node honestly addresses all the queries contained in the packet and sends it back to the IDS node. If any node provides false information in the status packet, that node is classified as a malicious node. However, the approach is

not applicable in large-size networks, because the restricted mobility of nodes is the main drawback of this approach, as their speed cannot exceed 1 m/sec [7]. Rani et al. [23] proposed a swarm-based Artificial Bee Colony (ABC) technique against BHA and GHA. In this approach, a node that satisfies the ABC fitness function is considered a normal node. Otherwise, the node is considered a malicious node. However, the ABC technique is slow in obtaining accurate decisions resulting in high end-to-end delay.

Tripathi et al. [31] proposed an ECT (Entity-Centric Combined Trust) model based on cooperation among vehicles to detect packet drop attacks. In this model, the satisfaction level of a node is based on the closeness position, transition reliability, and contact frequency. The combination of direct and indirect trust is calculated which gives an exact idea about the behavior of a node. In this model, the periodic exchange of trust values among various nodes results in a huge routing overhead. Arun et al. [32] proposed the SBGM (Smart Black-hole and Gray-hole Mitigation) mechanism based on the Dynamic Time Warping (DTW) technique. The RSU is deployed in the promiscuous mode that observes the flow of packets from each vehicle. SBGM first calculates the Dropped Packet Time Series (DPT) of each node and then compares the two DPT series and discovers their DTW distances. If a pair of nodes with DTW is found greater than the dynamic distance threshold, the pair is added to the suspected nodes list. If the count of a node in the list exceeds the entire number of nodes in the RSU range, that node is declared as an attacker node. The main drawbacks of adopting DTW technique are the heavy computational overhead and time complexity require to find an optimal path in real-time.

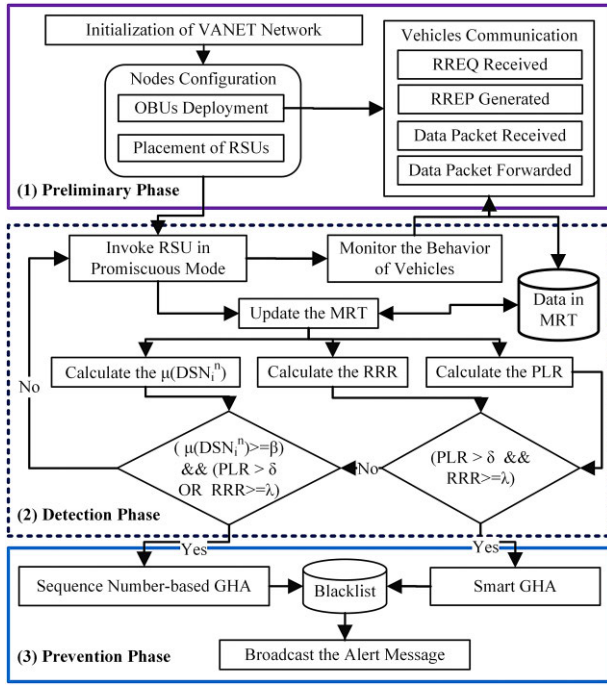


FIGURE 4. The framework of the proposed DPGHA.

Reviewing the relevant literature critically, it is evident that there are many limitations in the existing approaches. For example, the majority of these approaches [15], [30], [32] rely on additional IDS and control packets, which cause excessive routing overhead and delay. Similarly, approaches like [22], [29], [31] are dependent on clustering, trust value, and cryptographic techniques. The major drawbacks of these approaches are the generation of significant routing overhead due to the continuous formation of clusters and the exchange of heavy-weightage information among nodes. To address these challenges, a novel approach known as the Detection and Prevention of GHA (DPGHA) is proposed in this paper. The proposed approach is capable of detecting as well as preventing both varieties of GHA efficiently in terms of increased PDR and throughput, reduced routing overhead, and delay while achieving a maximum detection rate.

IV. THE PROPOSED DPGHA

In this section, the proposed DPGHA is presented in detail. The overall system design framework of the proposed DPGHA, its components, and operations are illustrated in Figure 4.

It consists of three main phases: the *preliminary phase*, the *detection phase*, and the *prevention phase*. Each phase has different operations and functionalities integrated to achieve security against GHA in VANET. First, we explain the preliminary phase, which contains the underlying system model of VANET, the node properties, the communication paradigm, the placement of RSUs, and some assumptions. Next, we present our novel detection and prevention phases in detail.

TABLE 1. Notations and abbreviations.

Symbol	Description
N	Node: a Vehicle or an RSU
A	Acceleration of a Vehicle
S	Source Node
D	Destination Node
E	Edge
V	Vehicle
R_T	Routing Table
T_R	Transmission Range
DSN	Destination Sequence Number
λ	Abnormal RREQ/RREP Threshold
δ	Packet Drop Threshold
$\mu(DSN_i^n)$	Mean Value of DSN of an Individual Vehicle
β	Mean Value of all DSN (vehicles) Threshold
$\dot{S}$	Speed of a Vehicle
t	Time
i and j	Variables: Start from 1, 2, 3, ..., n

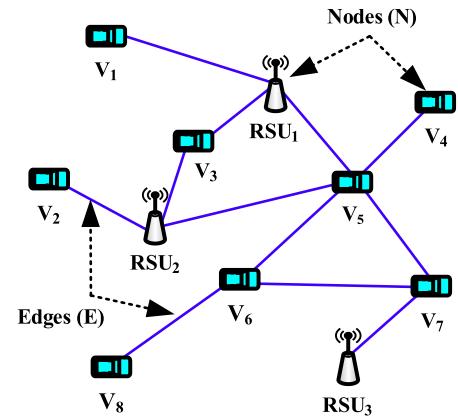


FIGURE 5. Graphical representation of a VANET.

A. PRELIMINARY PHASE

In this section, we provide the overall system model with some mathematical proof of concepts of node communications in VANET. The notations and abbreviations used in this paper are listed in Table 1.

A VANET is represented as a graph $G = (N, E)$, where N is a finite set of nodes (vehicles or RSUs) i.e., $N = \{x : x \in V_i \vee RSU_i\}$, and E is a finite set of edges connecting these nodes i.e., $E = \{\{x, y\} : x \in V_i \wedge y \in V_j \text{ and } (x \text{ and } y \in to G)\}$. The edges provide the communication links among nodes in the VANET, shown in Figure 5. In such a dynamic network the cardinality of nodes $|N|$ remains constant over a specific time while the cardinality of links $|E|$ can be changed due to the high mobility of nodes.

Let there is an X independent experiment with two possible outcomes (i.e., success or failure). The binomial distribution provides the probability of one successful outcome being P while the other is $1 - P$ (failure). Assume that there are N number of nodes and E number of edges in the network generated from the graph $G=(N, E)$ model. To measure the average number of edges $\langle E \rangle$, we need to model the probability that there will be exactly E edges out of the total number of node pairs $N(N-1)/2$. Let the maximum number of node

pairs is $E_{max} = N(N-1)/2$, each pair has a probability P to form an edge is calculated by the following binomial distribution equation (1).

$$P_E = \binom{E_{max}}{E} P^E (1-P)^{E_{max}-E} \quad (1)$$

The probability (P_k) that a node has exactly K edges is calculated by the following binomial distribution equation (2).

$$P_k = \binom{N-1}{k} P^k (1-P)^{N-1-k} \quad (2)$$

When a vehicle V_i is within the Transmission Range (T_R) of another vehicle V_j , i.e., $(d(V_i, V_j) \leq T_R)$, they are assumed to be connected by a dedicated unidirectional link $l_i \in E$. When vehicle V_i sends a packet, it can receive directly by vehicle V_j through a link l_i . A link $l_i = (V_i, V_j)$ (V_i, V_j) between two distinct vehicles exists if their Euclidean distance [33] is less than or equal to the T_R , i.e., $l_i = \{(V_i, V_j) | (POS_i - POS_j) \leq T_R\}$. Let V_A and V_B are two distinct vehicles at different positions in the T_R with speed S and time t_0 . The positions of V_A and V_B are (X_a, Y_a) and (X_b, Y_b) respectively. The Euclidean distance between V_A and V_B is calculated by (3).

$$d(V_i, V_j) = \sqrt{(X_b - X_a)^2 + (Y_b - Y_a)^2} \quad (3)$$

When the distance between two vehicles exceeds T_R , packets are transmitted indirectly between them in a multi-hop manner. Assume that (X_s, Y_s) and (X_n, Y_n) are the coordinates of a source vehicle V_s and its neighboring vehicle V_n , with their corresponding speeds S_s and S_n , transmission range T_R at any time t_0 . So, the lifetime of the link (l_i) between V_s and V_n is calculated by (4).

$$l_{V_s, V_n} = T_R - \frac{\sqrt{(X_n - X_s)^2 + (Y_n - Y_s)^2}}{V_s - V_n} \quad (4)$$

Let V_i and V_j are two vehicles and assume that they are traveling in the same direction. The position, speed, and acceleration of the vehicle V_i at time t_0 are (X_i, Y_i) , $\dot{S}_i$, and A_i respectively. The position, speed, and acceleration of the vehicle V_j at time t_0 are (X_j, Y_j) , $\dot{S}_j$ and A_j respectively. Assume that (X'_i, Y'_i) and (X'_j, Y'_j) are the two new positions of vehicles V_i and V_j at t_1 respectively as illustrated in Figure 6.

The distances traveled by both vehicles during the time $(t = t_1 - t_0)$ are represented by (5) and (6) [34].

$$X'_i - X_i = \frac{1}{2} A_i t^2 + \dot{S}_i t \quad (5)$$

$$X'_j - X_j = \frac{1}{2} A_j t^2 + \dot{S}_j t \quad (6)$$

From the above two equations, the time t can be calculated by the following quadratic formula in (7).

$$\frac{1}{2} (A_i - A_j) t^2 + (\dot{S}_i - \dot{S}_j) t + \{(X_i - X_j) - (X'_i - X'_j)\} = 0 \quad (7)$$

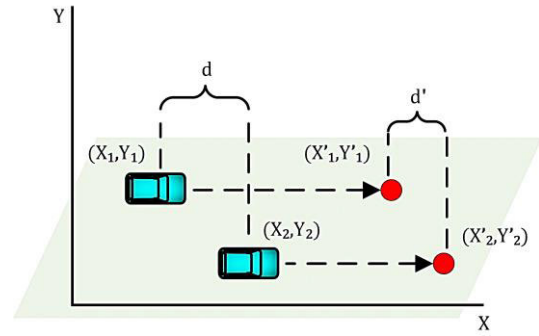


FIGURE 6. The positions of two vehicles along abscissa axes.

The term $(X_i - X_j)$ is the distance between the two vehicles at a time t_0 whereas $(X'_i - X'_j)$ is the distance between the two vehicles at the time t_1 . These two distances can be denoted by d and d' respectively. If vehicle V_i and vehicle V_j have the same acceleration, then the time that the vehicles stay in direct communication is calculated by (8) and (9).

$$(\dot{S}_i - \dot{S}_j) t + (d - d') = 0 \quad (8)$$

$$t = \frac{(d' - d)}{(\dot{S}_i - \dot{S}_j)} \quad (9)$$

where $\dot{S}_i \neq \dot{S}_j$ and $|d'| \approx T_R$. Assume that the accelerations of V_i and V_j are not the same, then the time t in which the two vehicles are able to communicate can be calculated from equation (7) by the quadratic formula in (10).

$$t = \frac{-|\dot{S}_i - \dot{S}_j| + \sqrt{(\dot{S}_i - \dot{S}_j)^2 - 2(A_i - A_j)(d - d')}}{(A_i - A_j)} \quad (10)$$

Let vehicle V_i and vehicle V_j are traveling in the opposite direction along the abscissa axes. The distances of the two vehicles are given in equations (5) and (6). From these two equations the time t at which the distance between vehicles V_i and V_j on the x-axis will be $|X'_i - X'_j|$ is represented by (11).

$$\frac{1}{2} (A_i - A_j) t^2 + (\dot{S}_i + \dot{S}_j) t + d - d' = 0 \quad (11)$$

The maximum time t for which the two vehicles can be in direct communication ($|d'| \approx T_R$) is calculated by (12).

$$t = \begin{cases} \frac{d' - d}{\dot{S}_i + \dot{S}_j}, & \text{if } A_i = A_j \\ \frac{-(\dot{S}_i + \dot{S}_j) + \sqrt{\Delta}}{(A_i + A_j)}, & \text{otherwise} \end{cases} \quad (12)$$

RSUs play an important role in the proposed DPGHA since they are fixed as roadside devices with substantial storage and computing capabilities. Each RSU is invoked in promiscuous mode, which operates as an IDS node, observing and sniffing all data and control packets sent and received by its neighbors and storing them in the Master Routing Table (MRT). The proposed DPGHA handles both types of GHA (i.e.,

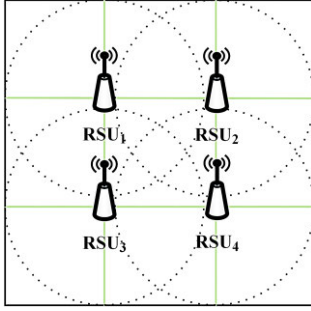


FIGURE 7. Placement of RSUs.

Smart GHA and Sequence Number-based GHA) with the help of three different parameters, including Packet Loss Ratio (PLR), Ratio of RREQ and RREP (RRR), and mean values of sequence numbers. These parameters are based on the malicious properties of the gray-hole nodes listed in Section II. To improve the performance and avoid heavy routing overhead, minor modifications have been made here to the default operation of the AODV routing protocol.

Assumptions- In order to build a logical framework for the development of the proposed DPGHA, it is necessary to make some prior assumptions to carry out its operations. These assumptions are rational and beneficial for the design consideration of VANETs, which are as follows:

- i. The proposed scheme requires the presence of at least one RSU, where two neighbor RSUs are required to be located within each other transmission range in order to exchange messages smoothly, as depicted in Figure 7.
- ii. Each RSU is invoked in promiscuous mode, allowing it to observe all incoming and outgoing traffic within its transmission range.
- iii. All the nodes should be exclusively identifiable, i.e., either as gray-hole or genuine nodes.
- iv. In addition to the routing table, each vehicle maintains a blacklist table, whereas an RSU keeps both the MRT and the blacklist tables.
- v. All nodes are in good condition and they must participate in the process of route discovery and route maintenance.

B. DETECTION PHASE

Keeping in mind the above system model and design framework for an AODV-based VANET, the detection phase of GHA begins. Initially, the RSU node is invoked in the promiscuous mode that periodically monitors all neighboring vehicles to determine whether various numbers of packets received by a node are passed to their next-hop nodes or not. To detect Smart GHA, an RSU counts the number of packets received, forwarded, and generated by each neighboring node and stores them in the MRT as shown in Table 2. The RSU first calculates the Packet Loss Rate (PLR) value of all the received and forwarded data packets of each node in the MRT

according to (13) and (14).

$$DP_D = \sum_{i=1}^n DP_R - \sum_{i=1}^n DP_F \quad (13)$$

$$PLR = \frac{DP_D}{\sum_{i=1}^n DP_R} * 100 \quad (14)$$

where DP_D denotes data packets dropped, DP_R denotes data packets received, and DP_F denotes data packets forwarded. If the PLR value of a vehicle is found to be greater than the threshold (δ) i.e. $PLR(V_k) > \delta$, then the RSU marks it as a suspicious vehicle. In this model, the value of threshold (δ) is set at 3%, because the standard AODV protocol for a normal node has a PDR of 97–98% [35]. Next, the RSU checks the amount of abnormal differentiation of the RREQ and RREP control packets of each vehicle and then compares it with the threshold value (λ). If an intermediate vehicle is not the destination itself and never forwards an RREQ packet for a given route, but instead responds with an RREP packet, its suspicious value is recorded in the MRT. To determine the rate of abnormal differences in control packets the RSU node calculates the Ratio of RREQ received and RREP generated (RRR) according to (15).

$$RRR = \frac{\sum_{i=1}^n RREP_G}{\sum_{i=1}^n RREQ_R} * 100 \quad (15)$$

where $RREQ_R > 0$. If a vehicle's RRR value is found greater than or equal to the threshold value (λ) i.e., $RRR \geq \lambda^{th}$, then the RSU marks it as a suspicious vehicle. An RSU declares a vehicle as a Smart GHA if its $RRR \geq \lambda^{th}$ and $PLR > \delta^{th}$, and then moves it to the blacklist. The maximum value for λ is assumed to be 70% in this model. This is obviously an abnormal indicator of generating a large number of RREPs in response to the RREQs differentiating the harmful property of a Smart GHA from the group of normal nodes.

Next, the RSU node checks the fake RREP with a high sequence number stored in the MRT. A Destination Sequence Number (DSN) is included in every RREP packet to identify the route's freshness. Despite having a fresh route to the destination, a Sequence Number-based GHA generates a fake RREP with a higher DSN to get involved in the route and attract the flow of traffic towards itself. To identify such a malicious node, an RSU computes the mean (μ) value of all the recorded RREPs' DSN exchanged by each node in its MRT table according to (16).

$$\mu(DSN_i^n) = \frac{\sum_{i=1}^n DSN_i^n(RREP_i^n(V_i))}{n} \quad (16)$$

where $\mu(DSN_i^n)$ is the mean value of the DSN of all the RREPs of an i^{th} vehicle recorded in the MRT and $i = 1, 2, 3, \dots, n$. After computing the $\mu(DSN_i^n)$ value, the RSU calculates a threshold value (β) according to (17).

$$\beta = \frac{\sum_{i=1}^n \mu(DSN_i^n)}{n} \quad (17)$$

where β is a dynamic threshold value that frequently changes each time a new process of GHA detection is initiated and n is the total number of vehicles recorded in the MRT. Now the

TABLE 2. Entry in MRT.

V_i^n	$\mu(DSN_i^n)$	$\Sigma(RREQ_R)$	$\Sigma(RREP_G)$	$\Sigma(DP_R)$	$\Sigma(DP_F)$	PD	PLR	RRR	$\mu(DSN_i^n) \geq \beta$	$PLR > \delta$	$RRR \geq \lambda$
V_1	21	40	15	550	545	5	0.909	37.500	No	No	No
V_2	25	60	25	920	910	10	1.086	41.666	No	No	No
V_3	23	50	46	850	810	40	4.705	92.000	No	Yes	Yes
V_4	18	45	12	640	630	10	1.562	26.666	No	No	No
V_5	200	80	75	1400	1370	30	2.142	93.750	Yes	No	Yes
V_6	32	65	40	960	895	65	6.770	61.538	No	Yes	No
V_7	60	41	25	1620	1590	30	1.851	60.975	Yes	No	No
V_8	17	39	19	210	205	5	2.380	48.717	No	No	No
$\beta = 49.5$											

RSU node compares each vehicle's μ^{th} value with β^{th} value. If a vehicle's μ^{th} value is found greater than or equal to the β^{th} value, then the RSU marks it as a suspicious vehicle. An RSU declares a vehicle as a Sequence Number-based GHA if its $\mu(DSN_i^n) \geq \beta^{th}$, and either its $PLR > \delta^{th}$ or its $RRR \geq \lambda^{th}$, and then moves it to the blacklist.

V_k (ID)

$$= \begin{cases} \text{SmartGHA}, & \text{if } (PLR > \delta^{th} \text{ AND } RRR \geq \lambda^{th}) \\ \text{Seq.No.GHA}, & \text{if } \left(\mu(DSN_i^n) \geq \beta^{th} \text{ AND } (PLR > \delta^{th} \text{ OR } RRR \geq \lambda^{th}) \right) \\ \text{Normal}, & \text{otherwise} \end{cases} \quad (18)$$

According to the proposed detection phase as in equation (18), a suspicious vehicle V_k is declared to be a Smart GHA and is moved into the blacklist if its $PLR > \delta^{th}$ and its $RRR \geq \lambda^{th}$. Similarly, a suspicious vehicle V_k is considered to be a Sequence Number-based GHA and moved to the blacklist, if its $\mu(DSN_i^n) \geq \beta^{th}$, and either its $PLR > \delta^{th}$ or its $RRR \geq \lambda^{th}$. Otherwise, a vehicle V_k is considered a normal node.

Experiment—Figure 8 and its corresponding Table 2 show the detection phase experimental scenario. It contains a total number of 8 vehicles under the observation of a single RSU. Here we assume that V_3 is a Smart GHA while V_5 is a Sequence Number-based GHA and all the rest of the vehicles are normal. As discussed above, the RSU is invoked in the promiscuous mode that observes and sniffs all data or control packets exchanged by each neighboring node lying within its transmission range.

To detect Smart GHA, an RSU counts the number of packets received, forwarded, and generated by each neighboring node and stores them in the MRT table as shown in Table 2. The RSU first calculates the PLR value for each V_1 to V_8 is calculated according to equations (13) and (14) in the MRT. Next, the RSU node compares the PLR value of each vehicle with the threshold value δ (equal to 3%). As shown in Table 2, the PLR values of both vehicles V_3 and V_6 are greater than δ . Hence, the RSU marks them as suspicious vehicles. Next, an RSU node checks the ratio of abnormal

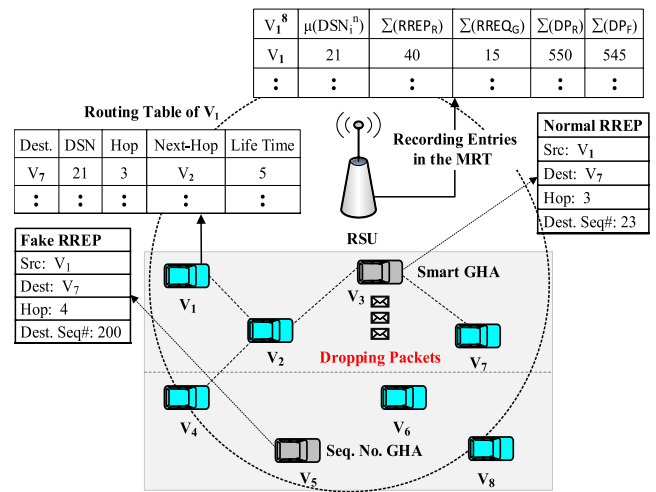


FIGURE 8. An experimental scenario of DPGHA.

differences of control packets (RRR value) exchanged by each node according to equation (15). Then RSU compares the RRR value of each vehicle with the threshold value λ (equal to 70%).

It can be seen in Table 2 that the RRR values of V_3 and V_5 vehicles are greater than λ . According to equation (18), V_3 is declared a Smart GHA and moved to the blacklist because its PLR is greater than δ and its RRR is greater than λ . Although V_5 is still considered a suspicious vehicle. Because packets may be lost due to other reasons such as link breakage, medium contention, or transmission queue overflow.

To detect Sequence Number-based GHA, first, the RSU computes the mean value i.e., $\mu(DSN_i^n)$ of all the recorded RREPs' DSN disseminated by each node according to equation (16). After computing $\mu(DSN_i^n)$, the RSU calculates a threshold value (β) according to equation (17), shown in Table 2 ($\beta = 49.5$). It can be seen in Table 2 that the $\mu(DSN_i^n)$ values of V_5 and V_7 vehicles are greater than β value. Thus, according to equation (18), V_5 is declared as a Sequence Number-based GHA and moved to the blacklist, because its $\mu(DSN_i^n)$ value is greater than β and its RRR value is greater

than λ^{th} . Algorithm 1 illustrates the complete process of the detection phase.

Algorithm 1 Detection of GHA

Input: Vehicle, RSU, RREQ_R, RREP_G, DP_R, DP_F, PLR, RRR, DSN
Output: Detection of GHA; Allows genuine nodes to communicate

- Initialization:** $i = 0, 1, 2, 3, \dots, n$
- Set RSU in Promiscuous Mode
- Monitors $\rightarrow$
 $RREQ_{R_i}^n, RREP_{F_i}^n, DP_R, DP_F, DSN_i^n(RREP_i^n)(V_i^n)$
- MRT $\leftarrow$
 $RREQ_{R_i}^n, RREP_{F_i}^n, DP_R, DP_F, DSN_i^n(RREP_i^n)(V_i^n)$
- MRT $\leftarrow$ Calculate $DP_D(V_i^n) = \sum_{i=1}^n DP_R - \sum_{i=1}^n DP_F$
- MRT $\leftarrow$ Calculate $PLR(V_i^n) = DP_D / \sum_{i=1}^n DP_R * 100$
- MRT $\leftarrow$ Calculate $RRR(V_i^n) = (\sum_{i=1}^n RREP_G \sum_{i=1}^n RREQ_R) * 100$
- if** $(PLR(V_k) > \delta^{th} \& \& RRR(V_k) \geq \lambda^{th})$
 - Type (1) $\leftarrow V_k(ID)$
 - Blacklist $\leftarrow$ Type(1)
 - goto** step25
- else**
 - goto** step15
- end**
- MRT $\leftarrow$ Calculate $\mu(DSN_i^n(V_i)) = \sum_{i=1}^n DSN_i^n(RREP_i^n(V_i)) / n$
- MRT $\leftarrow$ Calculate $\beta = \mu(\mu(DSN_i^n(V_i)))$
- if** $(\mu(DSN_i^n(V_k))) \geq \beta^{th} \text{ AND } (PLR > \delta^{th} \text{ OR } RRR \geq \lambda^{th})$
 - Type (2) $\leftarrow V_k(ID)$
 - Blacklist $\leftarrow$ Type(2)
 - goto** step25
- else**
 - MRT $\leftarrow V_k(ID)$
 - goto** step3
- end**
- RSU broadcasts Alert_{message} $V_k(ID) \rightarrow (V_i^n \text{ AND } RSU_i^n) \in T_R$

C. PREVENTION PHASE

Likewise, keeping in view the system model and network scenario of the VANET discussed earlier, here in this section we explain the proposed prevention phase. As in the detection phase, once an RSU detects a GHA (either Smart or Sequence Number-based GHA), it broadcasts the identity of that vehicle to all its neighboring nodes (RSUs or vehicles) via an alert message as illustrated in Figure 9.

The alert message contains the issuing identity of an RSU, the identity of the GHA, the type of GHA, and the time of identification. When a node (RSU or vehicle) receives an alarm message from an RSU, it first examines its blacklist table for a GHA entry. The alert message is discarded if an entry for the gray-hole node already exists; otherwise, the node's ID is added to the blacklist. Subsequently, the participation of the gray-hole node in the route discovery process is prevented by checking the blacklist table. A node ignores the RREP from another node if its identity is found

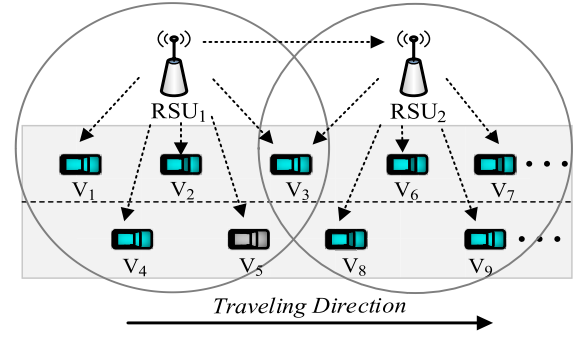


FIGURE 9. Broadcasting of an alert message.

TABLE 3. Blacklist table.

RSU' ID	GHA's ID	Type	Time
RSU ₁	V ₃	1	2022/01/05 10:12
RSU ₁	V ₅	2	2022/01/05 10:12

in the blacklist table. An entry of the blacklist table is shown in Table 3.

Here type-1 refers to Smart GHA while type-2 refers to Sequence Number-based GHA. Algorithm 2 illustrates the complete process of the prevention phase.

Algorithm 2 Prevention of GHA

Input: Vehicle, RSU, Blacklist, RREQ, RREP
Output: Prevention of GHA; Only genuine nodes can communicate

- Initialization:** $i = 0, 1, 2, 3, \dots, n$
- $(V_i^n \text{ v } RSU_i^n) \leftarrow$ Alert_{message} $(V_k(ID))$
- V_x searches $V_k(ID)$ in its Blacklist
- if** $(V_k(ID) == (Blacklist[V_k(ID)]))$
 - Discards** Alert_{message} $(V_k(ID))$
- else**
 - Blacklist $\leftarrow V_k(ID)$
- end**
- $(V_i^n \text{ v } RSU_i^n) \leftarrow RREQ \text{ v } RREP$
- if** $((RREQ \text{ v } RREP) == (Blacklist[V_k(ID)]))$
 - Discards** $(RREQ \text{ v } RREP)$
- else**
 - $(RREQ \text{ v } RREP) \rightarrow (V_i^n \text{ v } RSU_i^n)$
- end**
- end**

For normal vehicles- The implementation of the proposed DPGHA model for normal vehicles is mostly the same as the AODV protocol, except that:

- To keep track of gray-hole nodes, a blacklist table is employed in addition to the routing table.
- When an RSU sends an alert message, it first looks in its blacklist table for a gray-hole node. The alert message is discarded if an entry for the gray-hole node already exists; otherwise, the node's ID is added to the blacklist.



FIGURE 10. Map of Islamabad city.

- During route discovery time, a vehicle drops the RREQ or RREP packet if its ID is found in the blacklist table.
- For RSU nodes - The implementation of the proposed DPGHA mechanism for the RSUs is slightly different from the AODV protocol.
- The RSU employs two tables, the MRT as shown in Table 2 and the blacklist shown in Table 3. The MRT table keeps the records of all packets received, forwarded, and generated and the RREPs’ DSNs of each one-hop node. While the blacklist table is used to keep the record of gray-hole nodes.
 - When an RSU sends an alert message, it first looks in its blacklist table for a gray-hole node. The alert message is discarded if an entry for the gray-hole node already exists; otherwise, the node’s ID is added to the blacklist, and the alert message is rebroadcasted to all of its one-hop vehicles.
 - If the RREQ or RREP packet’s ID is detected in the blacklist table during route discovery, the RSU drops it.

V. IMPLEMENTATION AND PERFORMANCE ANALYSIS

We analyzed the performance of the proposed DPGHA approach using NS-2.35 [36] and SUMO-1.11 [37] simulators under Ubuntu-20.04 and the results are compared with the benchmark approaches i.e. AODV [11], ES-AODV [22], DDBG [30], AODV-PDBM [38], and SBGM [32].

Simulation Settings - To design a real-world traffic scenario of an urban environment, the “.osm” file of Srinagar Highway in Islamabad Capital City, Pakistan, is downloaded from the OpenStreetMap home page [39], shown in Figure 10. Mobility trace files for the respective “.osm” file are generated using the SUMO trace exporter, as shown in Figure 11. The resulting files are then loaded into NS-2 to analyze the performance of DPGHA. Afterward, the gray-hole nodes are injected into the model to provide a more realistic view of the network traffic.

We considered an area of 1000 × 1000 m² and ran the simulation for 900 seconds with various numbers of nodes ranging from 25 to 300 having the communication range of 250 m. Figure 12 shows one of the simulation experiment setups designed in NS-2 with a total number of 50 nodes, including 4 gray-hole (red) and 4 RSU nodes (blue).

The data obtained from the simulation trace file is the average of 10 random network traffic scenarios. For each simulation, 8% of nodes are selected as gray-hole nodes and

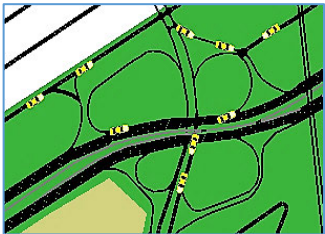


FIGURE 11. The SUMO GUI.

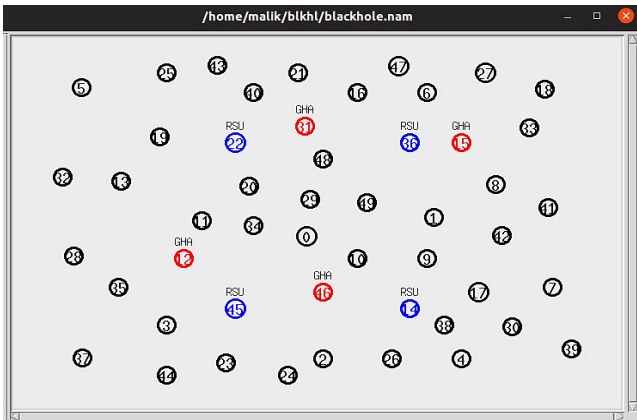


FIGURE 12. One of the initial states of a simulation experiment with fifty nodes.

TABLE 4. Simulation parameters.

Parameter	Value
Simulation Tools	NS-2.35, SUMO
Simulation Area	1000 m x 1000 m
Number of Nodes	25 min to 300 max
Simulation Time	900 sec
Vehicle Mobility	1 – 100 km/h
Base Routing Protocols	The AODV
Mobility Model	Urban: Random Way Point (RWP)
Standard Protocol	IEEE 802.11p
Gray-Hole Nodes	8%
Transport Protocol	UDP
Packet Size (bytes)	512b/s
Type of Traffic	CBR (Constant Bit Rate)
Antenna	Omni Directional
Communication Range	250 m

8% as RSU nodes. The simulation parameters are listed in Table 4.

Performances Evaluation Metrics - The performance of the proposed DPGHA approach is evaluated by comparing it with the existing benchmark approaches. For a fair comparison, the following performance evaluation metrics were used:

- Routing Overhead
- Packet Delivery Ratio
- Throughput
- End-to-End Delay
- Packet Loss Ratio
- Detection Ratio

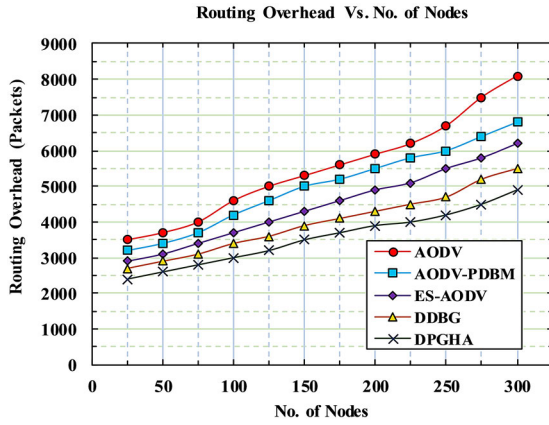


FIGURE 13. Comparison of routing overhead with an increasing number of nodes.

A. ROUTING OVERHEAD

Routing Overhead (ROH) refers to the total number of Control Packets (CP) transmitted to the total number of Data Packets (DP) received at the destination end. This metric shows how many CPs are required to send for a certain number of DPs. The main CPs of AODV routing protocol include RREQ, RREP, RERR (Route Error), and Hello message. ROH is calculated by (19).

$$ROH = \frac{\sum CP_{transmitted}}{\sum DP_{transmitted}} \quad (19)$$

Figure 13 shows the results of the routing overhead performance of the proposed DPGHA in comparison to benchmark approaches: AODV, AODV-PDBM, ES-AODV, and DDBG. The x-axis represents the number of nodes, while the y-axis represents the routing overhead (measured in packets). The routing overhead grows in proportion to the increase in the number of nodes. More RREPs are generated for the intended routes when there are more malicious nodes in the network, resulting in massive routing overhead. In the case of conventional AODV, more packets are lost due to the existence of more malicious nodes. As a result, the average routing overhead for conventional AODV is recorded as 5508.33.

The detection mechanism used in AODV-PDBM generates more control packets in terms of forwarding agent and backward agent packets, resulting in a significant routing overhead of 4983.33. However, the AODV-PDBM approach improves the average routing overhead of conventional AODV by 9.53%. The detection system of ES-AODV uses Elliptic Curve Cryptographic (ECC) based techniques which increase routing overhead, its average routing overhead is recorded as 4458.33. Similarly, in the case of DDBG, status packets are used to check the position of each node in the network.

However, the average routing overhead for DDBG is recorded as 3991.66, which is close to the proposed DPGHA. As shown in Figure 13, the proposed DPGHA outperforms and achieves the lowest possible routing overhead at the majority of points compared to the benchmark approaches. The routing overhead of the proposed DPGHA is recorded

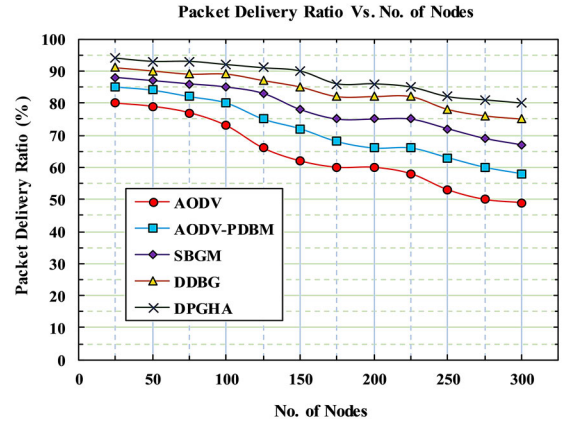


FIGURE 14. Comparison of PDR with an increasing number of nodes.

as 3558.33, which is the lowest average value among all the approaches. The reason for achieving minimum routing overhead is that the proposed DPGHA immediately blocks all the attacker nodes before causing harm to the network, through its threshold-based mechanism. In a nutshell, we found that our proposed DPGHA solution outperforms the existing approaches, having 35.40% less overhead than AODV, 28.59% than AODV-PDBM, 20.18% less than ES-AODV, and 10.85% than DDBG approach.

B. PACKET DELIVERY RATIO (PDR)

The PDR is referred to the ratio of the total number of data packets sent by the source node to the total number of data packets received by the destination node successfully, calculated by (20).

$$PDR = \left(\frac{\sum DP_{Received}}{\sum DP_{Sent}} \right) * 100 \quad (20)$$

Figure 14 shows the results of PDR for the proposed DPGHA in comparison to benchmark approaches. Following the graph, when the number of nodes increases in the network, the PDR decreases. The reason is that when the network size increases, the continual occurrence of packet collisions and packet drops increases due to the existence of more malicious nodes. The PDR for the conventional AODV declines due to a lack of security mechanisms for packet-dropping attacks.

The existence of GHA had a significant impact on AODV's performance; as the percentage of malicious nodes in the network increased, we recorded an average PDR performance of 63.91%. The security mechanisms used in the AODV-PDBM and SBGM approaches to deliver better performance results in terms of average PDR, were recorded as 71.58% and 78.33%, respectively. However, the average PDR for the DDBG approach is recorded as 83.83%, which is close to the proposed DPGHA. Because DDBG provides fast propagation of packets only among trustworthy nodes.

Similarly, the proposed DPGHA shows better performance in terms of PDR because the attacker nodes are promptly detected before causing harm to the network, using

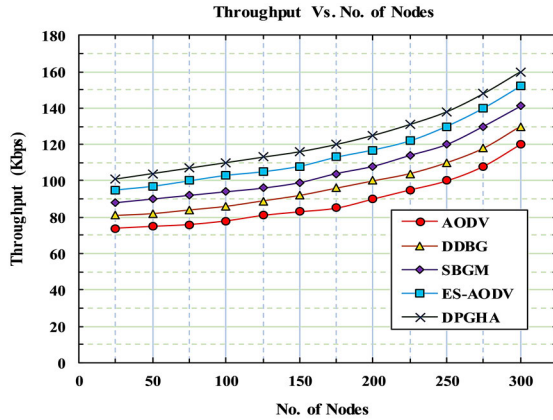


FIGURE 15. Comparison of throughput with an increasing number of nodes.

its threshold-based mechanism. After the detection of the attacker nodes, packets are securely and quickly delivered to their destinations. The average PDR for the proposed DPGHA is recorded as 87.75%, which is the highest value among all the approaches. The overall improvement of the proposed DPGHA in terms of PDR is increased by 37.28% to AODV and 22.58% to AODV-PDBM, 12.02% to SBGM, and 4.67% more PDR than the DDBG approach.

C. THROUGHPUT

The throughput metric is measured in bits per second and it refers to the actual data delivered in a given unit of time through the network as calculated in (21).

$$\text{Throughput} = \frac{\sum (\text{Packet}_{\text{Received}} * \text{PacketSize})}{\text{StopTime} - \text{StartTime}} \quad (21)$$

Figure 15 depicts the results of throughput (measured in Kbps) of the proposed DPGHA with a comparison to benchmark approaches. The existence of more malicious nodes, delay, link breakage, and congestion in the network has a significant impact on network throughput performance. Since the loss of packets in the network must be retransmitted, which degrades the overall throughput performance. The network throughput of the conventional AODV has the lowest significant values at each point due to the existence of GHA. In the case of the DDBG and SBGM approaches, the average throughputs are recorded as 97.66 and 106.33, respectively, since both of these approaches employ some security mechanisms that detect GHA immediately so that they achieve better throughput.

As compared to other approaches, the ES-AODV approach gets better results in terms of throughput because of the early detection of malicious nodes using ECC techniques. The average throughput of the proposed DPGHA has the greatest value of all the approaches with 122.75 Kbps. Because the mitigation mechanism used in DPGHA prevents the network from gray-hole nodes promptly and informs the normal nodes through alert messages to stop communication with the attackers. The proposed DPGHA achieves a significantly

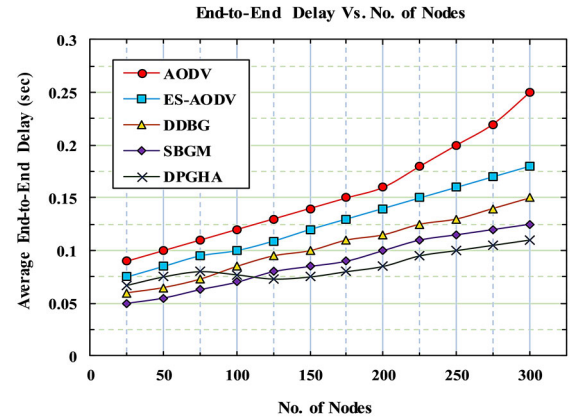


FIGURE 16. Comparison of E2E delay with an increasing number of nodes.

higher throughput than other protocols, with 38.30% higher than AODV, 25.68% higher than DDBG, 15.43% higher than SBGM, and 6.58% higher than the ES-AODV approach.

D. END-TO-END (E2E) DELAY

The E2E delay is the total time taken by a packet to reach a destination node successfully across a network. The lower the E2E delay indicates the better the model's performance. The E2E delay is calculated by (22).

$$\begin{aligned} E2EDelay &= \frac{\sum (\text{Time}(\text{Packet}_{\text{received}} - \text{Packet}_{\text{sent}})) * 1000 \text{ (ms)}}{\text{Total No. of Packets Delivered Successfully}} \quad (22) \end{aligned}$$

Figure 16 depicts the E2E delay (in seconds) for the DPGHA in comparison with the benchmark approaches. We observed that as the density of nodes increases in the network, the E2E delay also increases. Expanding a network's traffic density has a negative influence on E2E delay as more packets are lost due to the occurrence of link breakages, packet collisions, and packet-dropping attacks. The lost packets must be retransmitted, resulting in an increased E2E delay in the network.

In conventional AODV, the E2E delay is increased significantly due to a lack of security mechanisms for packet-dropping attacks. The average E2E delay for AODV is recorded as 0.154. Similarly, the average E2E delay for the ES-AODV and DDBG approaches is recorded as 0.126 and 0.104, respectively. The results of DDBG are better than the ES-AODV approach in terms of average E2E delay because the latter approach employs cryptographic-based techniques that require extra resource utilization. Compared to other approaches, the SBGM shows better performance in terms of E2E delay. Because it early detects malicious nodes through a time series analysis, which requires a minimum delay for its processing. Therefore, the average E2E delay of the SBGM approach is recorded at 0.088 seconds, which improves conventional AODV by 42.54%.

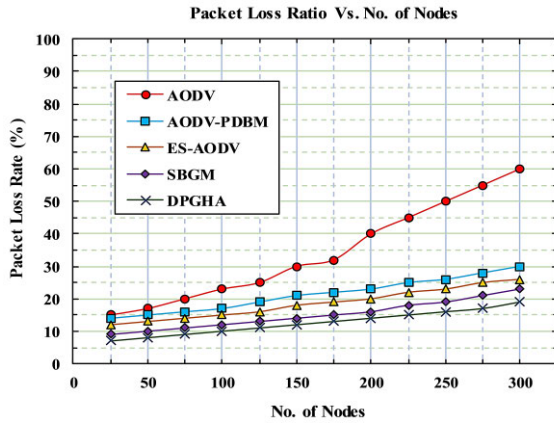


FIGURE 17. Comparison of PLR with an increasing number of nodes.

As depicted in the figure, the proposed DPGHA has the minimum average delay as compared to other approaches. Because the proposed DPGHA quickly detects the GHA from the network through the thresholds of abnormal differences among control and data packet mechanisms. As a result, the E2E delay performance for the proposed DPGHA is 0.085 seconds, which is the lowermost average value among all approaches. Hence, the improvement made by DPGHA in terms of E2E delay performance is reduced by 44.75% in comparison to the AODV, and results in a delay reduction of 3.85% in comparison to the SBGM approach.

E. PACKET LOSS RATE (PLR)

PLR simply represents the ratio of the number of lost packets to the total number of sent packets as calculated by (23). In VANETs, packets are lost due to gray-hole nodes in the network, link breakage, collision occurrence, etc.

$$PLR = \left(\frac{\sum DP_{Sent} - \sum DP_{Received}}{\sum DP_{Sent}} \right) * 100 \quad (23)$$

Figure 17 depicts the simulation results, indicating the PLR values in percentage on the y-axis while the number of nodes grows on the x-axis. As we can observe, the PLR goes up gradually against increasing the number of nodes. Because more packets are lost as a result of more malicious nodes, frequent occurrence of link breakages, and packet collisions. The rise of such factors has severe impacts on PLR.

As shown in Figure 17, the conventional AODV severely suffers from the above-mentioned factors, due to a lack of adequate security mechanisms. The ratio of PLR is consistently increased from 15% to 60% when the number of nodes increases from 25 to 300. However, the PLR for AODV-PDBM is lower than the AODV at each point. Because the AODV-PDBM approach employs a security mechanism that detects GHAs.

Similarly, ES-AODV and SBGM approach achieve relatively better performance as they detect gray-hole nodes early in the network. The average PLR for the ES-AODV approach is recorded as 18.58%, while for SBGM it is

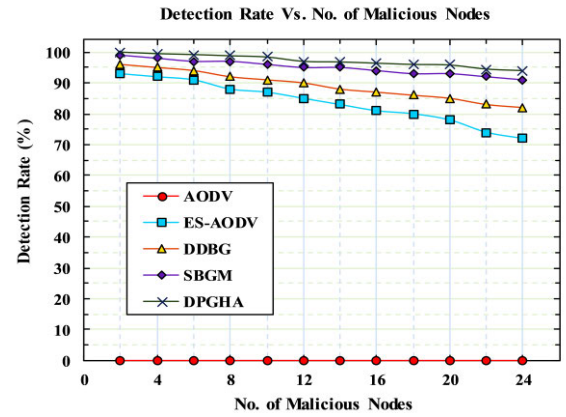


FIGURE 18. Comparison of detection rate with an increasing number of nodes.

15.08%, which is close to the proposed DPGHA. The SBGM shows better performance in terms of PLR metrics. Because it early detects malicious nodes through time series analysis techniques, resulting in minimum packet loss. Similarly, the proposed DPGHA uses thresholds of abnormal differences among control/data packet approaches to quickly exclude gray-hole nodes from the network and offer a secure path for packet delivery. Therefore, the overall performance of the proposed DPGHA in terms of PLR is reduced by 16.57% compared to the SBGM approach, which has the lowest PLR performance.

F. DETECTION RATE

An important metric for evaluating the accuracy and effectiveness of any IDS is the detection rate. The detection rate is sometimes called the sensitivity or True Positive Rate (TPR). It measures the correct identification of the actual attacker/gray-hole nodes. TPR, or detection rate, is calculated by (24).

$$TPR = \left(\frac{TP}{TP + FN} \right) * 100 \quad (24)$$

where TP is True Positive, and FN is False Negative. The FN measures the number of attacker nodes that are identified as normal nodes. Figure 18 and its corresponding Table 5 depict the results of a statistical comparison of the proposed DPGHA's detection rate to the benchmark approaches.

In Figure 18, the number of malicious nodes on the x-axis is plotted against the detection rate in terms of percentage on the y-axis. We observed that as the number of malicious nodes in the network grows, the detection rate declines. Because the presence of more malicious nodes results in more packet-dropping attacks. Moreover, as shown in the Figure, the conventional AODV severely suffers from packet drop attacks because it has been designed without taking into account the security aspects, thus its detection rate is recorded as 00.0%. Similarly, in the case of the ES-AODV approach, the average detection rate is recorded as 83.66%.

TABLE 5. Simulation results of statistical analysis of the detection rate.

Normal Nodes	Malicious Nodes	TPR for AODV	TPR for ES-AODV	TPR for DDBG	TPR for SBGM	TPR for DPGHA
25	2	00.0%	93.2%	96.0%	99.2%	100%
50	4	00.0%	92.0%	95.2%	98.8%	99.5%
75	6	00.0%	91.8%	94.0%	97.8%	99.2%
100	8	00.0%	88.4%	92.0%	97.0%	98.8%
125	10	00.0%	87.0%	91.0%	96.2%	98.4%
150	12	00.0%	85.8%	90.2%	95.8%	97.0%
175	14	00.0%	83.4%	88.2%	95.2%	96.8%
200	16	00.0%	81.8%	87.2%	94.4%	96.4%
225	18	00.0%	80.0%	86.0%	93.4%	96.0%
250	20	00.0%	78.8%	85.8%	93.0%	95.8%
275	22	00.0%	74.4%	83.0%	92.8%	94.5%
300	24	00.0%	72.8%	82.2%	91.0%	94.0%

The techniques used in the DDBG approach receive replies from every normal node, whereas malicious nodes do not reply properly or drop packets. Thus, the DDBG approach has a lower chance of detecting a malicious node, with a recorded detection rate of 89.08%. Finally, in the case of the SBGM approach, the average detection rate is recorded as 95%, which is 6.64% higher than the DDBG approach. The main reason for the better performance of the SBGM approach in terms of detection rate is that when the number of nodes is increased, the mean distance value becomes more accurate, allowing the approach to easily differentiate between honest and malicious nodes.

Contrary to existing approaches, the proposed DPGHA solution has the best performance with the highest detection rate, that is, an average of 97.2%. This higher detection rate is achieved by the proposed DPGHA because it quickly detects the hypocritical behavior of Smart GHA and Sequence Number-based GHA with the help of generating the threshold values of abnormal differences among data/control packets and sequence numbers of RREPs. Hence, the overall performance of the proposed DPGHA in terms of detection rate is increased by 2.3% compared to the SBGM approach, which has the highest detection rate of 95%.

The IoV and V2X will be vital themes in future ITSs. In this context, the event-driven tools will be advantageous in terms of reliability, real-time compression and computing complexity diminishing [40], [41]. Future research can be conducted to assess the feasibility of using these tools with the recommended methodology.

VI. CONCLUSION AND FUTURE WORK

A GHA in VANETs is a type of security attack in which a malicious node exhibits unpredictable behavior. It sometimes acts as an honest node, but later it starts misbehaving by partially, selectively, or completely dropping packets. In this work, we presented a new and efficient approach for the detection and prevention of gray-hole attacks called DPGHA to improve the overall security and performance of AODV-based VANETs. The approach relies on multiple thresholds of abnormal differences among received, forwarded, and generated control/data packets and the sequence number of RREPs.

DPGHA detects both types of GHA, namely Smart GHA and Sequence Number-based GHA. A comparison in terms of various performance evaluation metrics of the proposed DPGHA with the most relevant benchmark approaches is carried out by implementing and evaluating it in the NS-2 and SUMO simulators. The findings showed that the proposed DPGHA performs better than the benchmark approaches in terms of reducing routing overhead by 10.85%, increasing PDR by 4.67%, increasing throughput by 6.58%, decreasing end-to-end delay by 3.85%, and finally achieving a maximum detection rate of 2.3%.

Our work has several potential future extensions, including the implementation of DPGHA for autonomous and self-driving cars. Rapid advancements in vehicular technology pose new design and development challenges for existing security solutions in VANETs, making it important to explore new approaches to address security threats. Additionally, the proposed approach can be extended to geographic-based routing protocols, which have recently gained attention in many research studies in VANETs.

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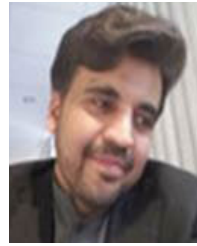
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